

A Positive Answer to the Integral-Kernel Question in Remark 24

arXiv automatic math run fa_banach_001, lane 0

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Status

Full solution, likely valid. This packet gives an affirmative answer to the natural question at the end of Remark 24 of Coine, Le Merdy, and Sukochev [1].

Source Question

The source paper is arXiv:1706.01662, *When do triple operator integrals take value in the trace class?*. The question appears in Remark 24 on page 38 of the arXiv PDF.

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the fact that $\phi \in L^\infty_\sigma(\lambda_B; \Gamma_2(L^1(\lambda_A), L^\infty(\lambda_C)))$. Hence according to (56), the conditions of Theorem 21 are also equivalent to

$$(57) \quad \phi \in L^\infty_\sigma(\lambda_B; \mathcal{I}(L^1(\lambda_A), L^\infty(\lambda_C))).$$

It is a natural question whether this implies the existence of a probability measure space (Σ, ν) and two functions $a \in L^\infty(\lambda_A \times \lambda_B \times \nu)$ and $b \in L^\infty(\lambda_B \times \lambda_C \times \nu)$ such that $\phi(t_1, t_2, t_3) = \int_\Sigma a(t_1, t_2, \omega)b(t_2, t_3, \omega) d\nu(\omega)$ for a.e. (t_1, t_2, t_3) . However we haven't been able to establish this yet.

We now turn to connections between Theorem 4 or Proposition 5 and the constructions of multiple operator integrals from [30] and [4].

Let $A_1, \dots, A_n$ be normal operators on a separable Hilbert space $\mathcal{H}$. Throughout we use the notations of Proposition 5. Let $(\Sigma, d\mu)$ be a σ -finite measure space and, for any $i = 1, \dots, n$, let

$$a_i : \Sigma \times \sigma(A_i) \longrightarrow \mathbb{C}$$

In the notation of the source, the question is whether

$$\phi \in L^\infty_\sigma(\lambda_B; \mathcal{I}(L^1(\lambda_A), L^\infty(\lambda_C)))$$

implies the existence of a probability space (Σ, ν) and functions

$$a \in L^\infty(\lambda_A \times \lambda_B \times \nu), \quad b \in L^\infty(\lambda_B \times \lambda_C \times \nu)$$

such that

$$\phi(t_1, t_2, t_3) = \int_\Sigma a(t_1, t_2, \omega)b(t_2, t_3, \omega) d\nu(\omega)$$

for almost every (t_1, t_2, t_3) .

Proof Intuition

For each fixed middle variable t_2 , the hypothesis says that the slice operator from $L^1(\lambda_A)$ to $L^\infty(\lambda_C)$ is integral, i.e. it is represented by a finite measure on a compact product of weak-star dual balls. The only extra issue is making these representing measures depend measurably on t_2 and using one common probability space for all slices.

The proof does exactly that. Hahn–Banach and the duality of Bochner L^1 -spaces give a bounded weak-star measurable kernel of finite measures. Since the compact space of dual-ball pairs is metrizable, that kernel can be randomized over $[0, 1]$. The Radon–Nikodym density of the signed kernel against the randomized probability kernel is absorbed into one scalar factor. The two coordinate projections of the randomized point give the two required bounded functions a and b .

A General Form

Throughout this section (X, μ) , (Y, η) , and (Z, ρ) are separable σ -finite measure spaces. Put

$$E = L^1(X, \mu), \quad F = L^1(Z, \rho).$$

For $\phi \in L^\infty(X \times Y \times Z)$, let $u_y : E \rightarrow F^*$ denote the slice operator

$$\langle u_y f, g \rangle = \int_{X \times Z} \phi(x, y, z) f(x) g(z) d\mu(x) d\rho(z).$$

Theorem 1. *Suppose*

$$y \mapsto u_y$$

belongs to $L^\infty_\sigma(Y; \mathcal{I}(E, F^))$. Then there is a probability space, which may be taken to be $([0, 1], m)$, and functions*

$$a \in L^\infty(X \times Y \times [0, 1]), \quad b \in L^\infty(Y \times Z \times [0, 1])$$

such that

$$\phi(x, y, z) = \int_0^1 a(x, y, s) b(y, z, s) dm(s)$$

for almost every (x, y, z) .

Moreover, if

$$M = \operatorname{ess\,sup}_{y \in Y} I(u_y),$$

then a, b can be chosen so that $\|a\|_\infty \|b\|_\infty \leq M$.

Proof. If $M = 0$, take $a = 0$ and $b = 1$. Assume $M > 0$.

Let

$$K = \mathbb{B}_{E^*} \times \mathbb{B}_{F^*},$$

where both dual balls carry their weak-star topologies. Since E and F are separable, K is compact metrizable. The injective tensor product

$$G = E \widehat{\otimes}_\varepsilon F$$

embeds isometrically into $C(K)$ by

$$J(f \otimes g)(\xi, \zeta) = \xi(f) \zeta(g).$$

The standard identification

$$G^* = \mathcal{I}(E, F^*)$$

is the one used in Remark 24 of the source paper; see Diestel–Uhl [2, Chapter VIII, Theorems 5 and 9].

The weak-star measurable essentially bounded field $y \mapsto u_y$ defines a bounded functional Λ on $L^1(Y; G)$ by

$$\Lambda(h) = \int_Y \langle u_y, h(y) \rangle d\eta(y).$$

Via the isometric embedding J , regard $L^1(Y; G)$ as a closed subspace of $L^1(Y; C(K))$. By Hahn–Banach, extend Λ to a functional $\tilde{\Lambda}$ on $L^1(Y; C(K))$ with $\|\tilde{\Lambda}\| = \|\Lambda\| \leq M$. The usual duality

$$L^1(Y; C(K))^* = L^\infty_\sigma(Y; M(K))$$

gives a weak-star measurable family $(\omega_y)_{y \in Y}$ of finite complex regular Borel measures on K , with

$$\operatorname{ess\,sup}_{y \in Y} \|\omega_y\| \leq M,$$

such that for every $h \in L^1(Y; C(K))$,

$$\tilde{\Lambda}(h) = \int_Y \int_K h(y, k) d\omega_y(k) d\eta(y).$$

In particular, for almost every y and every $f \in E$, $g \in F$,

$$\langle u_y f, g \rangle = \int_K \xi(f) \zeta(g) d\omega_y(\xi, \zeta). \quad (1)$$

We now put these representing measures on one probability space. Choose a base point $k_0 \in K$, and define a probability kernel

$$\pi_y = \frac{|\omega_y|}{M} + \left(1 - \frac{\|\omega_y\|}{M}\right) \delta_{k_0}.$$

Let $\gamma_y = d\omega_y/d\pi_y$. Then $|\gamma_y| \leq M$ for almost every y . Since K is compact metrizable, the standard randomization lemma for probability kernels on standard Borel spaces gives a measurable map

$$r : Y \times [0, 1] \longrightarrow K$$

such that $r(y, \cdot)_* m = \pi_y$ for almost every y . Write

$$r(y, s) = (\xi_{y,s}, \zeta_{y,s})$$

and set

$$\theta(y, s) = \gamma_y(r(y, s)).$$

Then $\theta \in L^\infty(Y \times [0, 1])$, $\|\theta\|_\infty \leq M$, and for every $h \in C(K)$,

$$\int_K h(k) d\omega_y(k) = \int_0^1 \theta(y, s) h(r(y, s)) dm(s) \quad (2)$$

for almost every y .

The maps $(y, s) \mapsto \xi_{y,s} \in \mathbb{B}_{E^*} = \mathbb{B}_{L^\infty(X)}$ and $(y, s) \mapsto \zeta_{y,s} \in \mathbb{B}_{F^*} = \mathbb{B}_{L^\infty(Z)}$ are weak-star measurable and bounded. By the standard identification of weak-star measurable L^∞ -valued maps with scalar L^∞ -functions, choose

$$\alpha \in L^\infty(X \times Y \times [0, 1]), \quad \beta \in L^\infty(Y \times Z \times [0, 1])$$

representing them, i.e.

$$\int_X \alpha(x, y, s) f(x) d\mu(x) = \xi_{y,s}(f), \quad \int_Z \beta(y, z, s) g(z) d\rho(z) = \zeta_{y,s}(g)$$

for all $f \in E$, $g \in F$, almost every (y, s) . Put

$$a(x, y, s) = \theta(y, s) \alpha(x, y, s), \quad b(y, z, s) = \beta(y, z, s).$$

Then $\|a\|_\infty \|b\|_\infty \leq M$.

Combining (1) and (2), for all $f \in E$ and $g \in F$, and for almost every y ,

$$\begin{aligned} \int_{X \times Z} \phi(x, y, z) f(x) g(z) d\mu(x) d\rho(z) &= \int_0^1 \theta(y, s) \xi_{y,s}(f) \zeta_{y,s}(g) dm(s) \\ &= \int_{X \times Z} \int_0^1 a(x, y, s) b(y, z, s) dm(s) f(x) g(z) d\mu(x) d\rho(z). \end{aligned}$$

Since simple tensors $f \otimes g$ span a dense subspace of $L^1(X \times Z)$, this identifies the two $L^\infty(X \times Z)$ -functions for almost every y . Equivalently,

$$\phi(x, y, z) = \int_0^1 a(x, y, s) b(y, z, s) dm(s)$$

for almost every (x, y, z) , as required. $\square$

Answer to Remark 24

Corollary 1. *The natural question in Remark 24 of [1] has an affirmative answer. Whenever*

$$\phi \in L^\infty_\sigma(\lambda_B; \mathcal{I}(L^1(\lambda_A), L^\infty(\lambda_C))),$$

there are a probability space (Σ, ν) and functions

$$a \in L^\infty(\lambda_A \times \lambda_B \times \nu), \quad b \in L^\infty(\lambda_B \times \lambda_C \times \nu)$$

such that

$$\phi(t_1, t_2, t_3) = \int_\Sigma a(t_1, t_2, \omega) b(t_2, t_3, \omega) d\nu(\omega)$$

for almost every (t_1, t_2, t_3) .

Proof. Apply Theorem 1 with

$$(X, \mu) = (\sigma(A), \lambda_A), \quad (Y, \eta) = (\sigma(B), \lambda_B), \quad (Z, \rho) = (\sigma(C), \lambda_C).$$

The scalar spectral measure spaces in the source paper are separable in the sense used there, because the normal operators act on a separable Hilbert space. $\square$

Verification Notes

The proof uses three standard pieces of measure-theoretic functional analysis: the representation of integral operators as the dual of the injective tensor product, Hahn–Banach extension from $L^1(Y; G)$ into $L^1(Y; C(K))$, and the randomization of probability kernels on compact metrizable spaces. No computational verification is involved.

The source-question crop was generated by:

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conda run --no-capture-output -n sandbox python code/make_problem_crop.py.
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Novelty Check

Local indexes were searched for arXiv:1706.01662, triple operator integrals, integral operators, L_σ^∞ , $I(L^1, L^\infty)$, Grothendieck, Peller, and the exact phrase from the source question. No existing packet for this question was found.

External web searches on June 16, 2026, used exact phrases including “It is a natural question whether this implies”, “1706.01662 natural question probability measure space”, and variants involving L_σ^∞ and $I(L^1, L^\infty)$. They did not locate a later paper explicitly answering the question. The novelty confidence is therefore bounded but not exhaustive.

Human Review Recommendation

Reviewers should focus on the measurable-kernel step: the Hahn–Banach extension gives a weak-star measurable family of measures on the compact product of dual balls, and the probability-kernel randomization must preserve the L^∞ -class identifications used for the two scalar factors. If that standard measurable-selection machinery is accepted, the answer is complete.

References

- [1] C. Coine, C. Le Merdy, and F. Sukochev, *When do triple operator integrals take value in the trace class?*, arXiv:1706.01662, to appear in Annales de l’Institut Fourier.
- [2] J. Diestel and J. J. Uhl, *Vector Measures*, Mathematical Surveys 15, American Mathematical Society, 1979.
- [3] O. Kallenberg, *Foundations of Modern Probability*, 2nd ed., Springer, 2002.